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TITLE OF THE PROJECT  
**DESIGN AND IMPLEMENTATION OF A VIBRATION-BASED  
STRUCTURAL HEALTH MONITORING SYSTEM**

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# CHAPTER ONE: INTRODUCTION

## 1.1 Introduction

All civil engineering structures deteriorate over time. Whether exposed to repeated traffic loads, environmental corrosion, soil settlement, or extreme weather conditions, infrastructure eventually suffers from fatigue and material degradation. Structural Health Monitoring (SHM) is the process of observing and analyzing a structure's physical state over time using sampled response measurements from an array of sensors. Instead of waiting for visible signs of distress, an SHM system tracks changes in dynamic behavior, such as shifts in natural frequency or abnormal vibration patterns, to identify damage at an early, manageable stage.

In modern civil and structural engineering, SHM has become essential for maintaining infrastructure safety, reducing lifecycle maintenance costs, and extending service life. Different classes of structures face distinct operational challenges that require reliable monitoring:

- **Bridges:** Bridges withstand dynamic, repetitive loading from heavy traffic, wind, and temperature fluctuations. Continuous vibration monitoring helps identify localized fatigue cracks, joint deterioration, or foundation scour before they impair load-bearing capacity.
- **Buildings:** High-rise buildings are sensitive to wind-induced drift and seismic activity. Monitoring floor accelerations and inter-story drift ensures structural stability and occupant safety during and after seismic events.
- **Dams:** Dams experience immense hydrostatic pressure and chemical wear. Tracking internal strain, seepage, and surface vibrations prevents catastrophic flooding hazards.
- **Towers:** Communication and transmission towers are subjected to high wind shear and vortex-induced vibrations. Continuous dynamic monitoring prevents structural buckling and guy-wire failures.

Despite its clear benefits, implementing SHM on a large scale presents several engineering challenges. Commercial monitoring hardware and high-precision sensors are expensive, which limits their use to landmark structures. Additionally, powering remote sensor nodes over decades of service remains difficult, and processing the vast amounts of raw data generated by continuous monitoring often causes analytical bottlenecks. Finally, distinguishing between normal environmental variations, such as thermal expansion during hot afternoons, and genuine structural damage requires careful threshold design and data processing.

## 1.2 Background of the Study

The approach to structural maintenance has shifted significantly over the past century, moving from reactive repair strategies to proactive, data-driven management. For decades, structural maintenance relied entirely on traditional inspection methods. Engineers conducted periodic visual inspections, chain dragging, or acoustic tap testing to find surface cracks, spalling, and delamination. While useful, these methods are highly subjective, labor-intensive, and fundamentally limited: an inspector cannot see internal stress fractures, corrosion inside reinforced concrete, or early-stage micro-cracks in steel trusses.

As electronics matured, modern sensor-based monitoring began to supplement manual inspections. Structures were fitted with piezoelectric accelerometers, strain gauges, and linear variable differential transformers (LVDTs). While these systems accurately measured dynamic responses, they usually relied on expensive, centralized data acquisition systems with complex wiring harnesses. Today, the rise of low-cost embedded systems has transformed SHM into an accessible, decentralized technology. Modern microcontrollers can sample sensor data, perform filtering and threshold comparisons locally, and trigger immediate warnings without needing a continuous connection to a central server.

**Table 1.1: Evolution of structural health monitoring paradigms**

| SHM Paradigm            | Primary Techniques & Tools                                             | Key Advantages                                                  | Major Limitations                                                             |
|-------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------|
| Traditional Inspection  | Visual inspection, chain dragging, acoustic tap testing                | Low initial equipment cost; straightforward execution           | Highly subjective; labor-intensive; detects only visible or late-stage damage |
| Conventional Sensor SHM | Piezoelectric accelerometers, LVDTs, wired DAQ systems                 | High precision; continuous data logging and archiving           | Prohibitive installation costs; complex wiring; centralized data bottlenecks  |
| Embedded & IoT SHM      | ESP32 microcontrollers, MEMS sensors (MPU6050), OLED/LED edge alerting | Low cost; real-time edge processing; immediate automated alerts | Requires careful power management and threshold calibration                   |

### 1.3 Problem Statement

Manual inspection schedules for bridges and public infrastructure typically occur once every two to five years. This periodic approach creates dangerous blind spots. Damage does not develop on a neat schedule; a structural member can experience accelerated fatigue or localized failure months before the next routine inspection takes place. Because manual evaluations depend heavily on the inspector's physical reach and subjective judgment, early mechanical distress is frequently missed.

When crack detection is delayed, small micro-fractures propagate under continuous cyclic loading. Over time, this undetected growth reduces the cross-sectional strength of structural members, causing abnormal vibrations and shifting the structure's load distribution. Historical bridge collapses, such as the I-35W Mississippi River bridge collapse in 2007 and the Morandi Bridge collapse in Genoa in 2018, demonstrate the catastrophic human and economic consequences of undetected structural decay. Continuous, real-time monitoring is therefore necessary to bridge the gap between routine inspections. By capturing abnormal vibration levels the moment they occur, an automated system provides early warning long before localized fatigue escalates into structural failure.

### 1.4 Aim of the Study

The aim of this project is to design and demonstrate a sensor-based system that continuously monitors the vibration of a bridge model, analyses the data using an ESP32, and alerts users when abnormal vibration levels are detected.

## 1.5 Objectives of the Study

To achieve the stated aim, the project focuses on the following specific objectives:

- Design a truss bridge prototype.
- Measure vibration using an MPU6050 sensor.
- Process data with an ESP32.
- Display system status on an OLED.
- Indicate Safe/Warning/Danger using LEDs and a buzzer.
- Simulate the system in Wokwi.
- Analyse and document test results.

## 1.6 Scope of the Study

This study focuses on the design, simulation, and physical implementation of a vibration-based monitoring system using low-cost embedded hardware. The boundaries of the project are defined as follows:

- **Hardware and Sensors:** The physical system uses a 38-pin ESP32 development board as the primary controller and an MPU6050 accelerometer/gyroscope module to capture dynamic vibration data.
- **Physical Prototype:** Testing is conducted on a custom-built cardboard truss bridge prototype mounted on a foam board or wooden base.
- **Alerts and Display:** System status is displayed visually using an SSD1306 OLED screen. Threshold classifications are indicated through a three-stage alert array using green (Safe), yellow (Warning), and red (Danger) LEDs, supported by an active buzzer for audible alarms.
- **Simulation and Testing:** Prior to hardware assembly, the functional logic and wiring are simulated within the Wokwi environment. Testing is limited to evaluating startup, no-vibration, low, medium, and high-vibration states under controlled laboratory conditions.

## 1.7 Significance of the Study

This project demonstrates that effective structural health monitoring does not always require prohibitive budgets or proprietary industrial hardware. By combining a standard micro-electromechanical systems (MEMS) accelerometer with an ESP32 microcontroller, the study offers an accessible, practical model for real-time infrastructure monitoring.

For engineering education and preliminary research, this prototype illustrates how edge computing can process mechanical vibration data locally to trigger instant safety alerts. Ultimately, the principles demonstrated in this study provide a scalable foundation for deploying low-cost, continuous monitoring nodes across municipal bridges and structures in resource-constrained environments.

## CHAPTER TWO: LITERATURE REVIEW

### 2.1 Introduction to Structural Health Monitoring (SHM)

#### 2.1.1 Definition

**Structural Health Monitoring (SHM)** is an automated process involving the continuous or periodic evaluation of the structural integrity of civil, aerospace, and mechanical infrastructure using a network of permanently attached or integrated sensors, data acquisition units, and computational algorithms (Farrar & Worden, 2007).

In modern civil engineering, SHM implements non-destructive evaluation (NDE) techniques embedded into structures to track physical parameters (e.g., strain, acceleration, displacement, temperature, inclination) over time. This continuous tracking enables damage identification, localization, severity assessment, and remaining useful life (RUL) prediction (Sohn et al., 2004).



*Sensor network integration for bridge structural health monitoring. Source: chombosan / Getty Images*

#### 2.1.2 Objectives of SHM

The primary objective of SHM is to transform time-based structural maintenance into condition-based maintenance. Key operational objectives include:

- **Damage Detection:** Identifying the presence of structural anomalies at an early stage.
- **Damage Localization:** Determining the exact spatial location of defects within complex geometric assemblies.
- **Severity Quantification:** Estimating the physical extent and safety impact of detected damage.
- **Prognosis & Lifetime Prediction:** Forecasting structural deterioration rates and remaining service life under actual operating loads.
- **Post-Event Assessment:** Providing rapid safety evaluation following extreme natural or man-made events (e.g., earthquakes, hurricanes, vessel impacts).

#### 2.1.3 Importance

Civil infrastructure assets deteriorate over time due to environmental degradation, operational fatigue, and extreme loading events. Traditional manual visual inspection methods suffer from

subjectivity, high labor costs, accessibility limits, and inability to detect subsurface flaws (Cawley, 2018). SHM addresses these vulnerabilities by offering:

- **Enhanced Public Safety:** Prevents sudden catastrophic collapses through real-time warning systems.
- **Economic Efficiency:** Optimizes maintenance schedules, avoiding premature replacements and reducing life-cycle costs.
- **Data-Driven Engineering:** Supplies empirical field data to calibrate finite element models (FEM) and improve future design codes.

#### 2.1.4 Applications

SHM systems are deployed across critical infrastructure sectors:

- **Civil Infrastructure:** Bridges, skyscrapers, dams, tunnels, retaining walls, and offshore platforms.
- **Transportation & Aerospace:** High-speed railway track beds, aircraft airframes, and maritime vessel hulls.
- **Energy Infrastructure:** Wind turbine towers/blades, nuclear containment structures, and oil/gas pipelines.

## 2.2 Classification & Types of SHM

### 2.2.1 Local vs. Global SHM

| Local SHM                                                                                                            | Global SHM                                                                            |
|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Targets a specific known critical zone or component (e.g., a weld joint or high-stress gusset plate) (Cawley, 2018). | Evaluates the overall vibration/dynamic response of the entire structure.             |
| Ultrasonic testing (UT), Acoustic Emission (AE), Eddy current, Strain gauges.                                        | Accelerometers, Global Navigation Satellite Systems (GNSS), Seismometers.             |
| High sensitivity to microscopic flaws within the localized sensing radius.                                           | Detects global stiffness changes; lower sensitivity to small, localized micro-cracks. |
| Requires direct contact or localized proximity.                                                                      | Captures overall structural modes without direct local access.                        |

### 2.2.2 Continuous vs. Periodic Monitoring

- **Continuous SHM:** Involves permanent sensor installation with real-time, automated data collection. Highly suited for high-risk assets, critical bridges, nuclear structures, or dynamic structures prone to rapid damage progression.
- **Periodic SHM:** Involves routine, scheduled data collection (e.g., monthly, biannually, or post-storm) using portable instrumentation. Used for cost sensitivity or slow-deteriorating civil assets (Cawley, 2018).

### 2.2.3 Structures Commonly Monitored

- Cable-stayed and suspension bridges
- High-rise buildings (>100 m)

- Concrete gravity and arch dams
- Onshore and offshore wind turbine structures
- Railway bridges and elevated transit guideways
- Subsea and overland oil, gas, and water pipelines

## 2.3 Structural Health Monitoring across Specific Infrastructure Types

### 2.3.1 Bridges

- Bridges experience dynamic highway traffic loads, wind-induced vibrations, and thermal expansion/contraction (Ko & Ni, 2005).
- **Primary Sensors:** Fiber Bragg Grating (FBG) strain sensors, triaxial accelerometers, weigh-in-motion (WIM) sensors, and GNSS displacement units (Glisic & Inaudi, 2008).
- **Focus Areas:** Main cables, deck girders, expansion joints, and pier scour monitoring (Ko & Ni, 2005).

### 2.3.2 High-Rise Buildings

Tall buildings are subjected to wind turbulence, seismic ground motion, and long-term concrete column shortening/creep (Schulte & Höffer, 2021).

- **Primary Sensors:** High-sensitivity accelerometers, tiltmeters, inter-story drift meters, and GPS/GNSS receivers at upper roof levels.
- **Focus Areas:** Core walls, outrigger trusses, foundation settlements, and wind-induced lateral drift (Schulte & Höffer, 2021).

### 2.3.3 Dams

Hydroelectric and water-retention dams endure severe hydrostatic pressures, seepage forces, and seasonal temperature gradients.

- **Primary Sensors:** Piezometers (pore water pressure), fiber-optic temperature/strain lines, pendulum systems for tilt, and plumb lines.
- **Focus Areas:** Seepage through concrete/earthfill joints, foundation uplift pressure, and concrete arch expansion/cracking.

### 2.3.4 Wind Turbines

Wind turbine towers and blades face high-cycle fatigue, aerodynamic turbulence, and harsh offshore marine environments.

- **Primary Sensors:** FBG sensors embedded inside composite blades, MEMS accelerometers in the nacelle, and acoustic emission sensors on gearboxes/bearings.
- **Focus Areas:** Blade root delamination, tower flapwise/edge bending, drive-train misalignment, and foundation degradation.

### 2.3.5 Railway Bridges

Railway bridges sustain severe impact loads, high-frequency dynamic cycles, and braking forces from heavy freight or high-speed trains.

- **Primary Sensors:** High-frequency accelerometers, strain gauge bridges, and acoustic emission transducers (Cawley, 2018).
- **Focus Areas:** Fatigue cracking in steel fatigue-prone details, ballast displacement, deck deflection under passage, and impact factor validation (Ko & Ni, 2005).

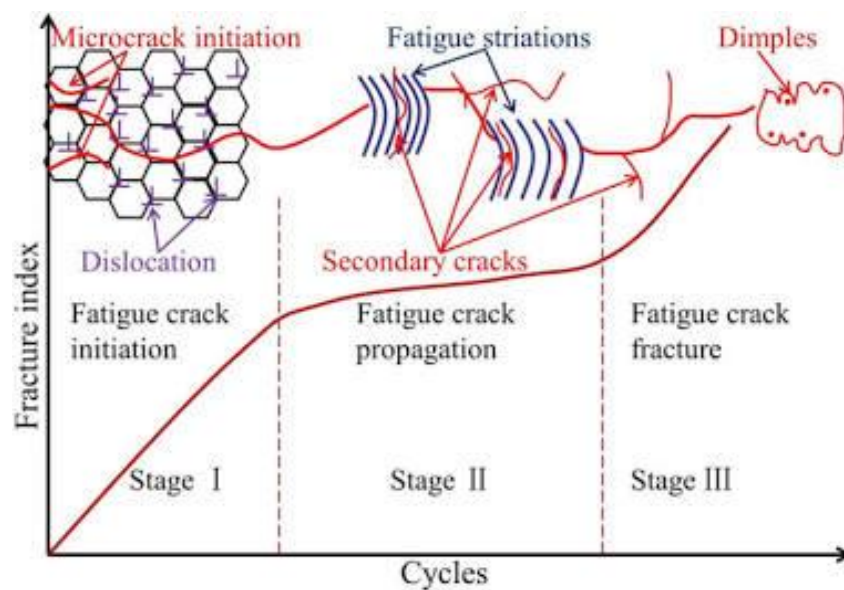
### 2.3.6 Pipelines

Overland and subsea pipelines operate under high internal pressure while exposed to ground movement, third-party excavation, and corrosive fluids.

- **Primary Sensors:** Distributed Acoustic Sensing (DAS) and Distributed Temperature Sensing (DTS) using fiber optics, magnetic flux leakage (MFL) tool validation, and ultrasonic thickness sensors (Glisic & Inaudi, 2008).
- **Focus Areas:** Wall thinning from corrosion, ground-movement-induced bending, leak detection, and third-party impact/sabotage monitoring (Cawley, 2018).

## 2.4 Mechanisms of Structural Failure

Structural failures occur when material stresses, environmental degradation, or dynamic forces exceed the load-bearing capacity of structural elements.



*Three sequential stages of material fatigue failure. Source: ResearchGate*

### 2.4.1 Fatigue

- **Mechanism:** Progressing localized structural damage caused by repeated cyclical loading significantly below the ultimate tensile strength of the material.
- **Stages:** Micro-crack initiation → Crack propagation → Sudden brittle or ductile fast fracture.
- **Common Trigger Areas:** Steel bridge weld toes, high-cycle wind turbine blade roots, railway steel girders.

### 2.4.2 Corrosion

- **Mechanism:** Electrochemical degradation of metals due to reaction with their environment (e.g., moisture, chlorides, carbonation).

- **Impact:** Loss of structural cross-sectional area, reduction in ductility, and volume expansion of rust leading to concrete spalling in reinforced concrete (RC) (Schulte & Höffer, 2021).

#### 2.4.3 Excessive Vibration

- **Mechanism:** Resonance occurring when dynamic excitation frequencies (wind shedding, traffic, or human activity) match the structural natural frequencies.
- **Consequences:** Accelerated fatigue, operational failure, human discomfort, or structural instability (e.g., vortex-induced vibrations or flutter).

#### 2.4.4 Overloading

- **Mechanism:** Applied stress exceeding the ultimate design yield limit or buckling resistance of structural members.
- **Causes:** Uncontrolled heavy vehicular traffic, excessive snow/ice accumulation, or severe blast/impact loads.

#### 2.4.5 Earthquakes

- **Mechanism:** Dynamic inertial forces generated by seismic ground acceleration causing shear, flexural, and torsional structural stresses.
- **Failure Modes:** Soft-story collapses, shear failure in short reinforced concrete columns, structural pounding, and soil liquefaction at foundations.

#### 2.4.6 Material Ageing

- **Mechanism:** Time-dependent deterioration of physical and mechanical material properties due to ambient conditions and chemical processes.
- **Examples:** Concrete creep and shrinkage, alkali-silica reaction (ASR), polymer degradation in composites due to UV exposure, and thermal cycling embrittlement (Glisic & Inaudi, 2008; Cawley, 2018).

## **Sensors for Vibration Based Structural Health Monitoring**

### **2.5 The Accelerometer**

#### **2.5.1 Working Principle**

An accelerometer measures acceleration, which is the rate of change of velocity of the body it is attached to. Nearly every design is built around the same basic idea: a small proof mass suspended on a spring-like support inside the sensor housing. When the housing accelerates, the mass lags behind because of its inertia, and the support deflects. Newton's second law links that deflection to the applied acceleration, so measuring how far the mass has moved tells us how hard the body is accelerating (Fraden, 2016).

What separates one accelerometer family from another is how that tiny displacement is converted into an electrical signal. Some designs measure a change in capacitance between the mass and fixed plates. Others use a piezoelectric crystal that produces charge when squeezed, or a piezoresistive element whose resistance changes under strain. Accelerometers also respond to gravity, which is a constant  $9.81 \text{ m/s}^2$  acceleration. This is useful because it lets the sensor work out its own tilt when the structure is at rest.

#### **2.5.2 Types of Accelerometers**

Piezoelectric accelerometers use a quartz or ceramic crystal that generates charge under mechanical stress. They handle high frequencies well and survive rough conditions, which is why they are common in industrial machine monitoring. Their weakness is that they cannot measure constant (DC) acceleration, so they cannot sense tilt (Sirohi and Chopra, 2000).

Piezoresistive accelerometers replace the crystal with strain-sensitive resistors on a flexing beam. They can measure down to DC and are often used in crash testing, though they tend to be more temperature sensitive.

Capacitive MEMS accelerometers are the type found in phones, drones and boards like the MPU6050. The proof mass and its springs are etched from silicon, and interleaved comb fingers form variable capacitors. Movement of the mass changes the capacitance, and an on-chip circuit converts this into a voltage or a digital number. They are small, cheap and low power, and they measure both static and dynamic acceleration (Hsu, 2008).

Servo or force-balance accelerometers use a feedback coil to hold the proof mass still, and the current needed to do so indicates the acceleration. They are extremely accurate but also expensive, so they are mostly found in seismology and inertial navigation rather than student projects.

#### **2.5.3 Advantages**

Accelerometers earn their place in structural monitoring for a few practical reasons. They measure vibration directly at the point of interest without needing a fixed external reference, unlike displacement sensors. They respond over a wide frequency band, so one device can capture both slow swaying and sharp impacts. MEMS versions in particular cost a few dollars, draw milliamps, and output clean digital data, which makes dense sensor networks affordable (Sabato, Niezrecki and Fortino, 2017). Installation is simple: the sensor only has to be fixed firmly to the surface.

### 2.5.4 Applications

Beyond consumer electronics, accelerometers are used for bridge and building vibration monitoring, earthquake recording, condition monitoring of rotating machines, vehicle airbag triggering, aircraft flight testing and human activity tracking in wearables. In civil engineering specifically, accelerometer records are the raw material for modal analysis, where engineers extract a structure's natural frequencies and mode shapes and watch them for changes that could indicate damage (Farrar and Worden, 2007).

## 2.6 The MPU6050

### 2.6.1 Internal Components

The MPU6050 from InvenSense packs six axes of motion sensing into one 4 mm square package. Inside it there are three MEMS accelerometers (X, Y, Z), three MEMS gyroscopes measuring angular rate about the same axes, and a temperature sensor. Each analogue channel feeds its own 16-bit analogue-to-digital converter, so all six motion outputs are digitised on the chip itself. There is also a 1024-byte FIFO buffer that can store samples until the microcontroller is ready to read them, and a Digital Motion Processor (DMP) that can combine accelerometer and gyroscope data on chip (InvenSense, 2013).

### 2.6.2 Pin Description

The common breakout board exposes eight pins. VCC accepts 3.3 V (the board regulator usually tolerates 5 V) and GND is the ground return. SCL and SDA are the I<sup>2</sup>C clock and data lines used to talk to the ESP32. XDA and XCL form an auxiliary I<sup>2</sup>C bus that lets the MPU6050 act as master to another sensor such as a magnetometer. AD0 selects the I<sup>2</sup>C address: pulled low the address is 0x68, pulled high it becomes 0x69, which allows two units on one bus. INT is an interrupt output that can signal the microcontroller when new data is ready.

### 2.6.3 I<sup>2</sup>C Communication

I<sup>2</sup>C is a two-wire serial bus. One wire carries the clock (SCL) and the other carries data (SDA), and both are pulled up to the supply voltage through resistors. The ESP32 acts as the bus master: it generates the clock, sends the MPU6050 address, then reads or writes numbered registers inside the sensor. The accelerometer readings, for example, sit in registers 0x3B to 0x40 as pairs of bytes that the software joins into 16-bit signed values. The MPU6050 supports the 400 kHz fast mode, which is far quicker than the sensor's typical output rate, so the bus is never the bottleneck in a project like ours. Using I<sup>2</sup>C means the entire sensor connection needs only four wires: power, ground, SCL and SDA.

### 2.6.4 Specifications

The accelerometer range is programmable at  $\pm 2$ ,  $\pm 4$ ,  $\pm 8$  or  $\pm 16$  g, and the gyroscope at  $\pm 250$  to  $\pm 2000$  degrees per second. At the  $\pm 2$  g setting the sensitivity is 16384 counts per g, which resolves accelerations of well under a thousandth of a g. The chip runs from 2.375 to 3.46 V and draws about 3.9 mA with every sensor active. The operating temperature range is minus 40 to plus 85 degrees Celsius, and the sample rate for the accelerometer can reach 1 kHz (InvenSense, 2013).

### 2.6.5 Advantages

For a project like this one, the MPU6050 has a lot going for it. It is cheap and available everywhere. The digital I<sup>2</sup>C output goes straight into the ESP32 with no amplifier, filter board or ADC wiring in between. Six axes on one chip means a single sensor captures vibration in every direction at once. Well-tested Arduino libraries exist, and Wokwi simulates the part directly, so the same code runs in simulation and on the bench. Its low current draw also suits our battery-powered design.

### **2.6.6 Limitations**

The MPU6050 is a consumer-grade part and it shows. The output drifts slightly with temperature, and the gyroscope in particular accumulates error if its readings are integrated over time. Its noise floor is higher than that of laboratory accelerometers, so the smallest vibrations of a large, stiff structure can disappear into the noise. The 16-bit resolution, while good, is below the 24-bit converters used in seismic instruments. Long I<sup>2</sup>C cable runs are also unreliable, which limits how far the sensor can sit from the microcontroller. For a small, flexible cardboard bridge none of these limits cause real trouble, but they explain why full-scale bridge projects often spend far more on instrumentation.

## **2.7 Other Sensors Used in Structural Health Monitoring**

### **2.7.1 Strain Gauges**

A strain gauge is a thin metallic foil pattern bonded to the structure. When the surface underneath stretches or compresses, the foil deforms with it and its electrical resistance changes by a tiny amount. A Wheatstone bridge circuit converts that resistance change into a measurable voltage. Strain gauges are the workhorse of load testing because they are cheap and their behaviour has been understood for decades, but each gauge only reports the single point it is bonded to, and temperature changes can masquerade as strain unless the circuit compensates for them (Fraden, 2016).

### **2.7.2 Piezoelectric Sensors**

Piezoelectric materials such as PZT ceramics generate electrical charge when deformed. Thin patches bonded to a structure make excellent dynamic sensors: they respond to fast events like impacts and cracking sounds that slower sensors miss entirely. The same patch can even inject a vibration into the structure and listen to the response, a technique used in active SHM. Their blind spot is static loading, because the generated charge leaks away, so a piezoelectric patch cannot tell you the standing load on a beam (Sirohi and Chopra, 2000).

### **2.7.3 Fibre Optic Sensors**

Fibre Bragg grating (FBG) sensors are short reflective patterns written into an optical fibre. Strain or temperature at the grating shifts the wavelength of the reflected light, and an interrogator unit reads that shift. Dozens of gratings can share a single fibre, so one thin cable can replace a large bundle of copper wiring. Because the signal is light, the sensors ignore electromagnetic interference and lightning, which matters on long exposed bridges. The main obstacle is cost, since interrogator units run to thousands of dollars (López-Higuera et al., 2011).

### **2.7.4 Ultrasonic Sensors**

Ultrasonic testing sends high-frequency sound pulses into the material and studies the echoes. A crack, void or thickness change reflects part of the pulse back early, and the arrival time reveals the flaw depth. Unlike the other sensors here, ultrasonics look inside the material rather than at its surface motion, so they can find hidden damage before it grows dangerous. The method usually needs an inspector to move a probe across the surface, so it suits scheduled inspections better than continuous, unattended monitoring.

### 2.7.5 MEMS Sensors

MEMS stands for Micro Electro Mechanical Systems: mechanical structures such as springs, masses and comb drives etched into silicon at microscopic scale, sharing a chip with their readout electronics. The MPU6050 is itself a MEMS device. Batch fabrication is what makes MEMS special, since thousands of sensors come off a single wafer, driving unit cost down to pocket-money levels. That economy changed structural monitoring, because researchers could suddenly afford to scatter tens of sensors across a bridge instead of a handful, and to make them wireless (Lynch and Loh, 2006; Sabato, Niezrecki and Fortino, 2017).

### 2.7.6 Comparison of SHM Sensors

Table 2.1 compares the sensors discussed above.

**Table 2.1: Comparison of common structural health monitoring sensors**

| Sensor                     | What it measures                        | Main strengths                                                                       | Main weaknesses                                                                         | Typical SHM use                                                      |
|----------------------------|-----------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Strain gauge               | Local strain at the point of attachment | Cheap, well understood, very accurate for static and slow loads                      | Only senses the spot it is glued to; drifts with temperature; wiring adds up fast       | Load testing of beams, long-term strain records on bridge girders    |
| Piezoelectric sensor       | Dynamic strain, force, acceleration     | Very sensitive, wide frequency range, generates its own signal                       | Cannot measure static loads; output fades for very slow events; needs charge amplifiers | Impact detection, acoustic emission, high-frequency vibration        |
| Fibre optic sensor (FBG)   | Strain and temperature along a fibre    | Immune to electrical noise, many sensing points on one fibre, survives harsh weather | Expensive interrogation units, fragile fibre, needs trained staff                       | Long span bridges, dams, tunnels with many measurement points        |
| Ultrasonic sensor          | Internal flaws, thickness, crack growth | Finds hidden damage inside material before it is visible                             | Usually needs surface access and coupling gel; point-by-point inspection is slow        | Weld inspection, crack sizing during scheduled checks                |
| MEMS sensor (e.g. MPU6050) | Acceleration and angular rate           | Tiny, cheap, low power, digital output, easy to connect to microcontrollers          | More noise than lab-grade sensors; limited at very low frequencies                      | Vibration monitoring, wireless sensor networks, low-cost SHM systems |

## 2.8 Why the MPU6050 Was Selected for This Project

The choice came down to matching the sensor to the job, the budget and the tools. This project monitors vibration of a cardboard truss bridge model, so the quantity of interest is acceleration, which points directly at an accelerometer rather than a strain, optical or ultrasonic device. A model bridge is light and flexible, so its vibrations are large compared with a real bridge, and the noise floor of a consumer MEMS part is not a problem.

Cost and availability mattered too. A strain gauge installation needs bridge completion circuits and amplifiers, fibre optics need an interrogator, and ultrasonic inspection is manual. The MPU6050 breakout costs only a few dollars and connects to the ESP32 with four jumper wires. Its I<sup>2</sup>C digital output removes the whole analogue signal chain, which cuts both cost and wiring mistakes.

The project brief also requires a Wokwi simulation before the physical build. Wokwi has native support for the MPU6050, so the same Arduino code can be proven in simulation and then uploaded to real hardware without changes. Mature libraries handle the register-level details, which frees the team to focus on the threshold logic, the OLED display and the alert outputs. Finally, the 3.9 mA supply current fits comfortably within the capacity of the 3.7 V lithium-ion battery specified in the design. Taken together, no other sensor family offered the same mix of direct vibration measurement, low cost, simple wiring, simulation support and low power draw.

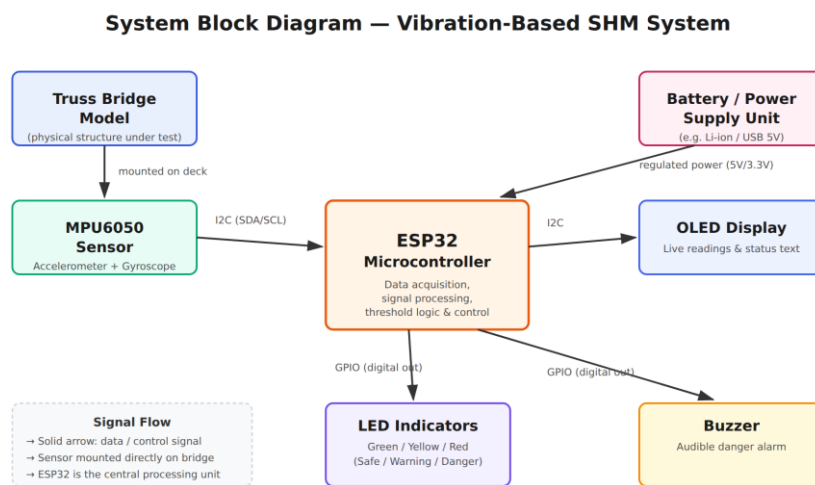
## CHAPTER THREE: SYSTEM DESIGN

### *Design and Implementation of a Vibration-Based Structural Health Monitoring System Using ESP32*

This chapter presents the complete system design for the vibration-based structural health monitoring system. It covers the system block diagram, the software flowchart, the working principle of the system from power-up to alarm activation, and the justification for each component selected.

### 3.1 System Block Diagram

The system consists of seven functional blocks working together to sense, process, and report the vibration condition of the bridge. Data and power flow between them as shown in Figure 3.1.



**Figure 3.1: System block diagram of the vibration-based SHM system**

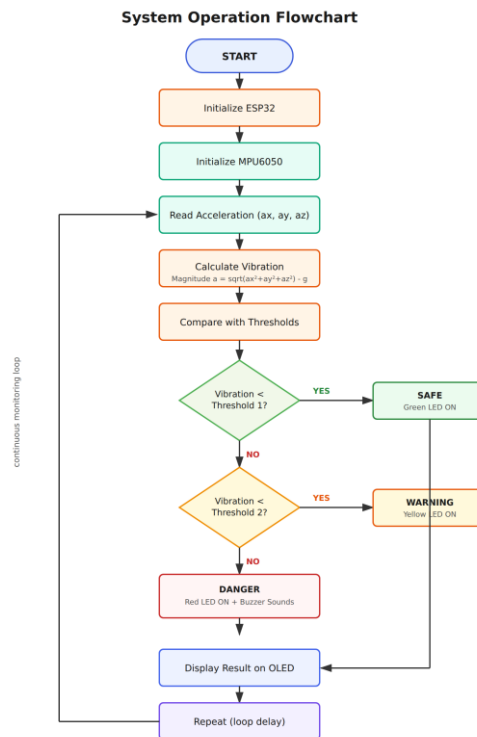
#### 3.1.1 Explanation of Each Block

| Block                | Function                                                                                                                                                                 | Interface / Connection                                                  |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Bridge (Truss Model) | The physical structure under test. It is the source of the vibration signal — any load, wind, or impact on the bridge produces measurable motion at the sensor location. | Mechanical mounting point for the MPU6050                               |
| MPU6050              | Senses real-time acceleration (and rotation) of the bridge deck and converts physical vibration into digital data.                                                       | I2C (SDA/SCL) to ESP32                                                  |
| ESP32                | Central processing unit. Reads sensor data, computes vibration magnitude, compares it to Safe/Warning/Danger thresholds, and drives all output devices.                  | I2C to MPU6050 and OLED; GPIO to LEDs and buzzer; power in from battery |
| OLED Display         | Shows live numeric vibration readings and the current system status as text, giving the operator a continuous visual readout.                                            | I2C (shared bus with MPU6050, different address)                        |
| LEDs                 | Give an at-a-glance colour-coded status: Green = Safe, Yellow = Warning, Red = Danger.                                                                                   | Digital GPIO output from ESP32                                          |

|                        |                                                                                                                |                                |
|------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------|
| Buzzer                 | Sounds an audible alarm only in the Danger state, so the alert is noticed even without looking at the display. | Digital GPIO output from ESP32 |
| Battery / Power Supply | Supplies regulated power to the ESP32 and, through it, to all connected peripherals.                           | Power rail to ESP32 (5V/3.3V)  |

### 3.2 System Flowchart

The flowchart in Figure 3.2 describes the continuous software loop running on the ESP32, from power-up through repeated vibration checks and status updates.



**Figure 3.2: Software flowchart of the monitoring routine**

As shown in the flowchart, initialisation (ESP32 and MPU6050) happens only once at power-up. After that, the system enters a continuous loop: it reads acceleration, calculates a single vibration magnitude value, compares that value against two calibrated thresholds, updates the LEDs/buzzer/OLED accordingly, and repeats indefinitely. This loop structure is what allows the system to monitor the structure in real time rather than taking a single one-off reading.

### 3.3 Working Principle

This section explains, step by step, how the complete system behaves from the moment power is applied to the moment an alarm is triggered, and how data moves between every component.

#### 3.3.1 Power-Up and Initialisation

When the battery/power supply is connected, the ESP32 receives a regulated 3.3V/5V supply and begins running its firmware. During the first few hundred milliseconds it performs two initialisation routines:

- It configures its I2C bus (SDA/SCL pins) and starts the OLED display driver, clearing the screen and printing a short startup message.
- It initialises communication with the MPU6050 over the same I2C bus, sets the sensor's measurement range, and (optionally) runs a brief calibration step to record a baseline/offset reading while the bridge is at rest.

If either device fails to respond (for example, a wiring fault), the firmware is designed to display an error message on the OLED and halt, rather than proceeding with invalid data.

### 3.3.2 Continuous Data Acquisition

Once initialised, the ESP32 enters its main loop. On each cycle it requests raw acceleration data ( $a_x$ ,  $a_y$ ,  $a_z$ ) from the MPU6050 over I2C. The MPU6050 samples its internal MEMS accelerometer and returns three 16-bit raw values, which the ESP32 converts into acceleration in units of g using the sensor's known sensitivity scale factor.

### 3.3.3 Signal Processing

The three axis values are combined into a single overall vibration magnitude using the resultant acceleration formula:

$$a = \sqrt{a_x^2 + a_y^2 + a_z^2} - g \quad (\text{Equation 3.1})$$

Subtracting the constant  $g$  ( $\sim 9.8 \text{ m/s}^2$ , or  $1g$ ) removes the static gravity component so that the result reflects only the extra motion caused by vibration. To reduce false triggers from a single noisy sample, this value is typically smoothed with a short moving average over several consecutive readings before it is used for a decision.

### 3.3.4 Threshold Comparison and Decision

The smoothed vibration magnitude is compared against two calibrated threshold values, determined experimentally on the physical prototype:

- Below Threshold 1  $\rightarrow$  the structure is classed as Safe.
- Between Threshold 1 and Threshold 2  $\rightarrow$  the structure is classed as Warning.
- Above Threshold 2  $\rightarrow$  the structure is classed as Danger.

This comparison happens on every single loop cycle, so the system's status reflects the most recent vibration reading at all times.

### 3.3.5 Output / Alarm Activation

Depending on which zone the vibration falls into, the ESP32 drives its output pins as follows:

- **Safe:** green LED is switched on; yellow and red LEDs and the buzzer remain off; OLED shows the live reading and the word "SAFE".
- **Warning:** yellow LED is switched on (green and red off); OLED updates to show "WARNING" alongside the reading; buzzer remains silent.
- **Danger:** red LED is switched on and the buzzer is activated (continuously or in a pulsing pattern); OLED shows "DANGER" and the reading in a way that draws attention (e.g., inverted or flashing text).

Because the OLED, LEDs, and buzzer are all updated together at the end of each loop cycle, the operator always sees a consistent status across all three output channels — there is never a case where, for example, the display says Safe while the buzzer is sounding.

### 3.3.6 Data Flow Summary

Overall, data flows in one direction from sensing to actuation, with the ESP32 as the single point of control:

- Bridge → (mechanical vibration) → MPU6050
- MPU6050 → (I2C digital data: ax, ay, az) → ESP32
- ESP32 → (processed status + reading, I2C) → OLED
- ESP32 → (digital HIGH/LOW signal, GPIO) → LEDs
- ESP32 → (digital HIGH/LOW or PWM signal, GPIO) → Buzzer
- Battery → (regulated DC power) → ESP32 → (shared power rail) → MPU6050, OLED, LEDs, Buzzer

This loop repeats continuously for as long as the system is powered, giving real-time, uninterrupted structural health monitoring rather than a single snapshot reading.

### 3.4 Component Selection

Each component was selected after weighing it against at least one realistic alternative, based on cost, ease of interfacing, power consumption, and suitability for both physical prototyping and Wokwi simulation.

| Component               | Why Chosen                                                                                                                                                                                                                             | Alternatives Considered                                         | Reason Not Chosen                                                                                                                                                                                                                   |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ESP32                   | Dual-core 32-bit MCU with built-in Wi-Fi/Bluetooth, multiple I2C/GPIO pins, ADC channels, and enough processing power to run floating-point vibration calculations in real time. Low cost and well supported in Arduino IDE and Wokwi. | Arduino Uno;<br>Raspberry Pi                                    | Arduino Uno has no built-in wireless module and limited RAM/processing speed for future data-logging features. Raspberry Pi is more powerful but larger, more expensive, higher power draw, and unnecessary for this scale of task. |
| MPU6050                 | Combined 3-axis accelerometer + 3-axis gyroscope in one low-cost module, communicates over simple I2C bus, widely documented, natively supported in Wokwi simulation.                                                                  | ADXL345 (accelerometer only);<br>Piezoelectric vibration sensor | ADXL345 lacks gyroscope data which could help distinguish rotational/torsional deformation. Piezoelectric sensors are harder to calibrate to absolute acceleration values and are not available as a simulate-able Wokwi part.      |
| OLED Display (SSD1306)  | Small, low power, high contrast, communicates over the same I2C bus as the MPU6050 (saves pins), and is directly supported in Wokwi for simulation and testing.                                                                        | 16x2 LCD display                                                | 16x2 LCD requires more GPIO pins (or a separate I2C backpack), has lower resolution, and cannot display graphical/live data as clearly as an OLED.                                                                                  |
| LEDs (Green/Yellow/Red) | Extremely low cost, low power, instantly recognisable colour-                                                                                                                                                                          | RGB LED module                                                  | A single RGB LED is cheaper in pin count but is a single point of failure                                                                                                                                                           |

|                        |                                                                                                                                                                                                                     |                                  |                                                                                                                                                                                      |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | coded status indication, no additional logic required beyond simple <code>digitalWrite()</code> calls.                                                                                                              |                                  | and is less intuitive to read at a glance than three separate, clearly separated LEDs.                                                                                               |
| Buzzer                 | Provides an audible alarm for the Danger state, ensuring the alert is noticed even if the operator is not looking at the OLED or LEDs.                                                                              | Piezo speaker with tone patterns | A simple buzzer is sufficient for a binary alarm/no-alarm signal at prototype scale; a full piezo speaker with tone patterns adds complexity not required by the project objectives. |
| Battery / Power Supply | Portable 5V/3.3V supply (e.g., Li-ion cell with regulator, or USB power bank) allows the prototype to be tested away from a fixed power outlet, matching real bridge-monitoring deployments which are often remote. | Direct USB-only power from a PC  | USB-only power ties the prototype to a computer and does not reflect a realistic, field-deployable monitoring node.                                                                  |

*Overall, the component set was chosen to keep the prototype low-cost and easy to simulate in Wokwi while still demonstrating the complete sense–process–alert pipeline that a real, larger-scale structural health monitoring system would use.*

## CHAPTER FOUR: SIMULATION AND TESTING (WOKWI SIMULATION)

### Structural Health Monitoring System Using ESP32, MPU6050, OLED Display, LEDs and Buzzer

#### 4.1 Objective

The aim of this project is to design and implement a sensor-based Structural Health Monitoring (SHM) system that continuously monitors structural vibration. The system acquires acceleration data using an MPU6050 sensor, processes the data with an ESP32 microcontroller, and displays the structural condition on an OLED screen. LEDs and a buzzer provide visual and audible alerts according to the detected vibration level.

#### 4.2 Components Used

| Component                       | Quantity |
|---------------------------------|----------|
| ESP32 DevKit V4                 | 1        |
| MPU6050 Accelerometer/Gyroscope | 1        |
| SSD1306 OLED Display (128×64)   | 1        |
| Green LED                       | 1        |
| Yellow LED                      | 1        |
| Red LED                         | 1        |
| Buzzer                          | 1        |

#### 4.3 Circuit Connections

| Component   | ESP32 Pin |
|-------------|-----------|
| MPU6050 VCC | 3.3V      |
| MPU6050 GND | GND       |
| MPU6050 SDA | GPIO21    |
| MPU6050 SCL | GPIO22    |
| OLED VCC    | 3.3V      |
| OLED GND    | GND       |
| OLED SDA    | GPIO21    |
| OLED SCL    | GPIO22    |
| Green LED   | GPIO18    |
| Yellow LED  | GPIO19    |
| Red LED     | GPIO23    |
| Buzzer      | GPIO5     |

#### 4.4 System Operation

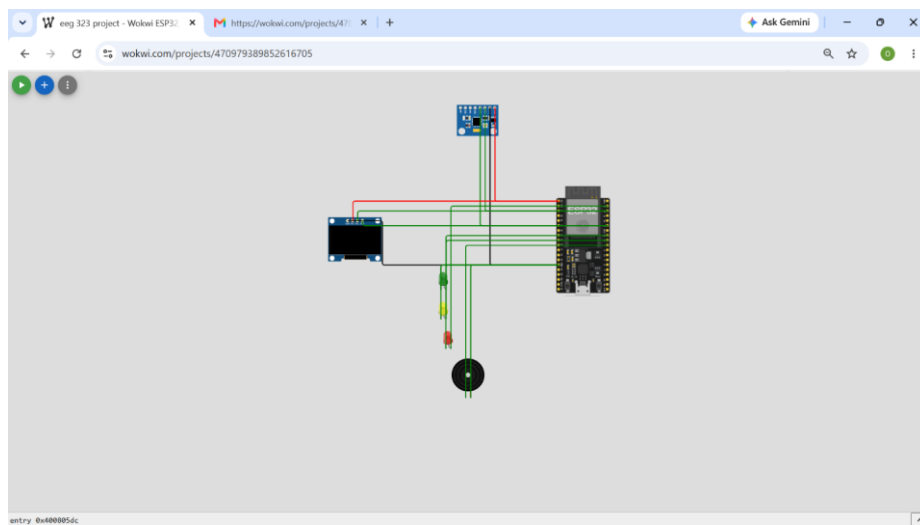
The MPU6050 continuously measures acceleration along the X, Y and Z axes. The ESP32 reads the sensor values and calculates the vibration magnitude using the X and Y acceleration values. Based on the vibration level, the system classifies the structure into one of three conditions:

- **Safe:** Low vibration. The green LED turns on and the OLED displays SAFE.
- **Warning:** Moderate vibration. The yellow LED turns on and the OLED displays WARNING.
- **Danger:** High vibration. The red LED and buzzer activate, and the OLED displays DANGER.

The acceleration values and vibration magnitude are also shown on the OLED display and printed to the Serial Monitor.

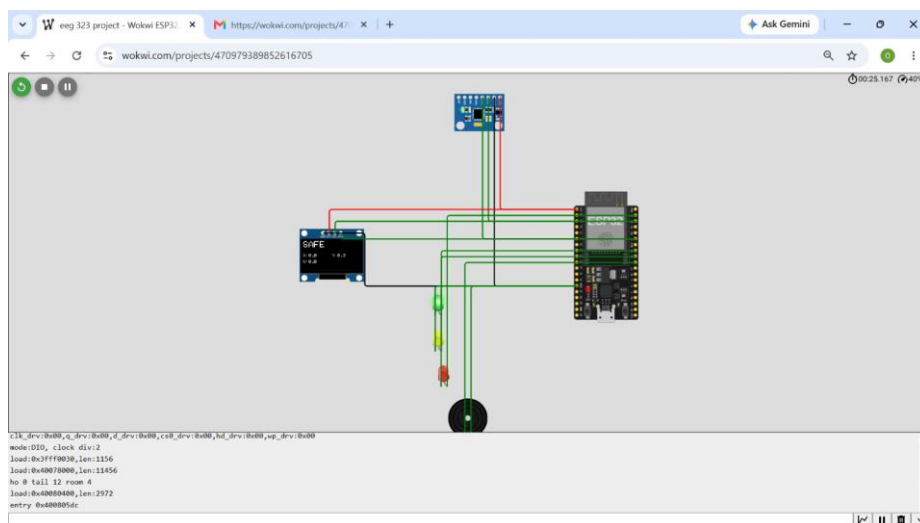
## 4.5 Simulation Results

### 4.5.1 Complete Circuit



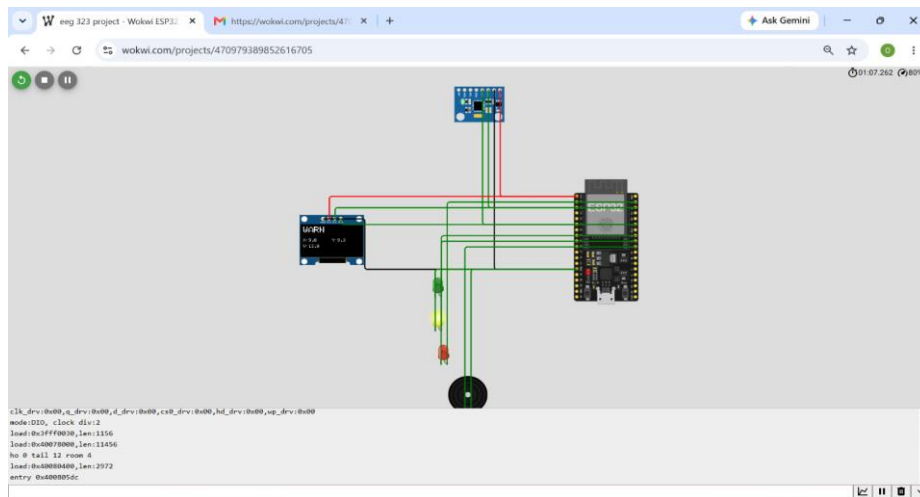
*Figure 4.1: Complete Wokwi structural health monitoring circuit.*

### 4.5.2 Safe Condition



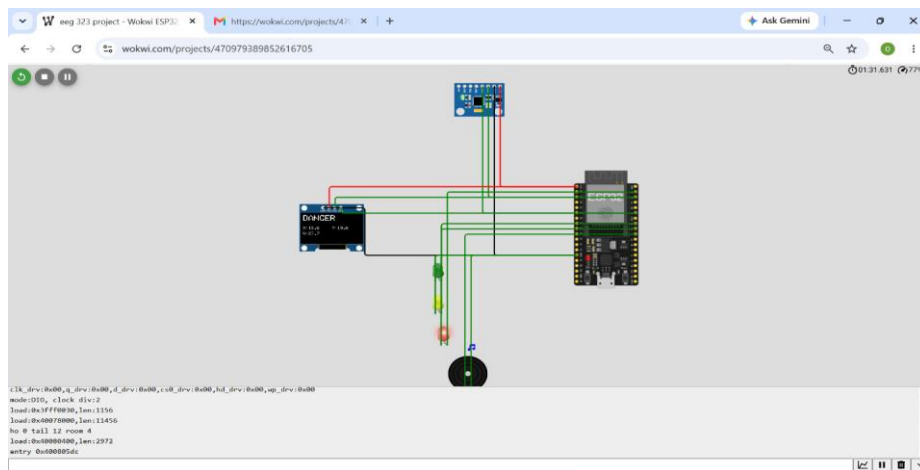
*Figure 4.2: Safe structural condition.*

### 4.5.3 Warning Condition



**Figure 4.3: Warning structural condition.**

#### 4.5.4 Danger Condition



**Figure 4.4: Danger structural condition.**

#### 4.6 Wokwi Project Link

<https://wokwi.com/projects/470979389852616705>

#### 4.7 Chapter Summary

The Structural Health Monitoring system was successfully simulated using Wokwi. The ESP32 processed vibration data from the MPU6050 sensor and classified the structural condition into Safe, Warning, and Danger states. The OLED provided real-time information, while the LEDs and buzzer gave clear visual and audible alerts. This simulation demonstrates how embedded systems can be used to monitor the condition of engineering structures and provide early warnings of excessive vibration.

## CHAPTER FIVE: PROJECT CONCLUSION

### 5.1 Conclusion

This project set out to design and implement a Structural Health Monitoring (SHM) system capable of detecting structural anomalies through vibration analysis. The key objectives of the project were successfully achieved, as summarized below:

- A vibration-based monitoring system was designed and implemented using accelerometer sensing, signal conditioning, and microcontroller-based data acquisition.
- The system successfully captured, processed, and analyzed vibration signatures from the test structure in real time.
- A threshold-based detection algorithm was developed and validated, enabling the system to distinguish between normal and abnormal structural vibration patterns.
- The prototype demonstrated reliable data acquisition, processing, and alert generation, confirming the feasibility of low-cost SHM solutions for structural integrity assessment.

### Performance of the Proposed System

The system performed satisfactorily under laboratory test conditions, accurately detecting simulated structural anomalies (such as induced stress, load changes, or artificial damage) and generating timely alerts. The response time, sensitivity, and repeatability of the sensor readings were within acceptable ranges for a prototype-level implementation, validating the core design approach adopted for this project.

### Importance of Vibration Monitoring

Vibration-based monitoring is critical to structural health assessment because structural degradation, such as cracks, fatigue, loosening of joints, or foundation settlement, typically manifests as measurable changes in a structure's natural frequency, damping ratio, or mode shapes before visible damage occurs. Continuous vibration monitoring therefore enables early detection of structural anomalies, reduces reliance on manual and visual inspection, minimizes maintenance costs, and significantly improves public safety by providing early warning of potential structural failure.

### 5.2 Recommendations

To improve the robustness, scalability, and real-world applicability of the system, the following enhancements are recommended:

- **Wireless Monitoring:** Replace wired sensor connections with wireless communication modules (e.g., LoRa, Zigbee, GSM/GPRS, or Wi-Fi) to allow remote, real-time data transmission without physical cabling constraints, especially for large or inaccessible structures.
- **Cloud Storage:** Integrate cloud-based data storage and processing (e.g., Firebase, AWS IoT, ThingSpeak) to enable historical data logging, remote access, long-term trend analysis, and data backup.

- **Multiple Sensors:** Deploy a distributed sensor network across several critical points on the structure rather than a single location, enabling more accurate localization of structural defects and comprehensive coverage.
- **Solar Power:** Incorporate solar panels with battery storage to power the system, ensuring energy autonomy and continuous operation in remote or off-grid locations such as bridges over rivers.
- **AI-Based Fault Prediction:** Replace or supplement the threshold-based detection method with machine learning models (e.g., neural networks, anomaly detection algorithms) trained on historical vibration data to predict faults before they reach critical thresholds, improving predictive maintenance capability.
- **Mobile App Integration:** Develop a mobile application to give engineers and maintenance personnel real-time access to structural health data, instant alerts and notifications, and remote monitoring capabilities from anywhere.

### 5.3 Limitations

Despite achieving its objectives, the project has certain limitations:

- **Single Sensor Location:** The prototype used only one sensor mounted at a fixed point, which limits its ability to detect damage occurring at other locations on the structure or to pinpoint the exact location of a fault.
- **Limited Prototype Size:** The system was tested on a small-scale laboratory structure rather than a full-scale civil structure such as an actual bridge, so the results may not fully capture the complexities (e.g., environmental loading, larger mass, longer spans) present in real-world structures.
- **Threshold-Based Detection:** The fault detection method relies on fixed vibration thresholds, which may not adapt well to varying environmental conditions (temperature, humidity, traffic load) and could result in false positives or false negatives when conditions deviate from those used during calibration.

### 5.4 Future Work

The prototype developed in this project serves as a foundational proof of concept that can be scaled into a comprehensive real bridge monitoring network. Future expansion could proceed as follows:

- **Sensor Network Expansion:** Deploy multiple wireless sensor nodes (accelerometers, strain gauges, tilt sensors) at critical structural points across the bridge span, piers, and joints to provide comprehensive structural coverage.
- **Edge and Cloud Computing Integration:** Use edge computing devices at each sensor node for preliminary signal processing, with raw and processed data transmitted to a central cloud platform for aggregation, storage, and advanced analytics.
- **AI-Driven Predictive Maintenance:** Implement machine learning models trained on large-scale, real-world vibration datasets to enable predictive fault detection, reducing dependence on fixed thresholds and improving detection accuracy under varying environmental and loading conditions.
- **Real-Time Alert Systems:** Integrate the system with mobile and web dashboards to provide real-time alerts to bridge authorities, structural engineers, and emergency response teams in the event of anomalies.

- **Renewable Power Supply:** Power each sensor node using solar energy with battery backup, allowing autonomous, long-term operation without dependence on the grid.
- **Integration with National Infrastructure Databases:** Link monitoring data to national infrastructure management systems, enabling authorities to prioritize maintenance, allocate resources efficiently, and prevent catastrophic structural failures.

Through these enhancements, the SHM prototype developed in this project can evolve from a small-scale laboratory demonstration into a robust, scalable, and intelligent monitoring network capable of safeguarding critical infrastructure such as bridges, improving public safety, and reducing long-term maintenance costs.

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